

Python Programming Assignment

Control Flow Structures: For Loops & While Loops

Course: Python Fundamentals

Total Problems: 23

Focus: Iteration & Logic Building

1. First N Natural Numbers

Loops

Write a program to print the first N natural numbers.

Input: 10 | **Output:** 1 2 3 4 5 6 7 8 9 10

2. First N Even Natural Numbers

Loops

Write a program to print the first N even natural numbers.

Input: 5 | **Output:** 2 4 6 8 10

3. First N Odd Natural Numbers

Loops

Write a program to print the first N odd natural numbers.

Input: 5 | **Output:** 1 3 5 7 9

4. First N Multiples of 3

Loops

Write a program to print the first N multiples of 3.

Input: 7 | **Output:** 3 6 9 12 15 18 21

5. First N Multiples of 5

Loops

Write a program to print the first N multiples of 5.

Input: 5 | **Output:** 5 10 15 20 25

6. Multiples of 2 Till N

Loops

Write a program to print all the multiples of 2 till N.

Input: 15 | **Output:** 2 4 6 8 10 12 14

7. Multiples of 2 or 3 Till N

Loops

Write a program to print all the numbers which are multiples of either 2 or 3 till N.

Input: 15 | Output: 2 3 4 6 8 9 10 12 14 15

8. Multiples of 2, 5 or 7 Till N

Loops

Write a program to print all the numbers which are multiples of either 2, 5 or 7 till N.

Input: 15 | Output: 2 4 5 6 7 8 10 12 14 15

9. First N Multiples of 3, 5 or 7

Loops

Write a program to print the first N multiples of either 3, 5 or 7.

Input: 10 | Output: 3 5 6 7 9 10 12 14 15 18

10. Sum of Digits

Loops

Find the sum of all digits in a positive integer.

Input: 123456789 | Output: 45

11. Count Number of Digits

Loops

Count the number of digits in a positive integer.

Input: 123456789 | Output: 9

12. Find Factors

Loops

Write a program to find factors of a given number.

Input: 20 | Output: 1 2 4 5 10 20

13. Count Factors

Loops

Write a program to count factors of a given number.

Input: 20 | Output: 6

14. Prime Number Check

Loops

Write a program to find whether the given number is a prime number or not.

Input: 11 | **Output:** Yes

15. Prime Numbers up to N

Loops

Print all prime numbers from 2 up to and including 'n'.

Input: 15 | **Output:** 2 3 5 7 11 13

16. Greatest Common Factor (GCF/GCD)

Loops

Write a program to find the greatest common factor of given 2 integers.

Input: 10 20 | **Output:** 10

17. Common Factors

Loops

Print the common factors of two positive integers n and m.

Input: 8 12 | **Output:** 1 2 4

18. Fibonacci Series

Loops

Write a program to get the Fibonacci series between 0 to 50.

19. Factorial Calculation

Loops

Write a program to find the factorial of a given number.

20. Sum of Even and Odd Numbers

Loops

Write a program that reads a set of integers, and then prints the sum of the even and odd integers.

21. Palindrome Number Check

Loops

Write a program to check whether the given number is palindrome or not. Palindrome is a number when reversed it gives the same number.

Example 1: Input = 121, Reverse = 121 (Palindrome)

Example 2: Input = 438, Reverse = 834 (Not Palindrome)

22. Armstrong Number Check

Loops

Write a program to print if a number is an Armstrong number or not.

Example: 153 is an Armstrong number because $1^3 + 5^3 + 3^3 = 153$.

23. Pattern Printing Practice

Nested Loops

Write Python programs using nested loops to print the following patterns:

Pattern A (Right-Angled Triangle):

```
*  
* *  
* * *  
* * * *  
* * * * *
```

Pattern B (Number Pyramid):

```
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5
```